Marblehead → Provincetown — Overnight Race Weather

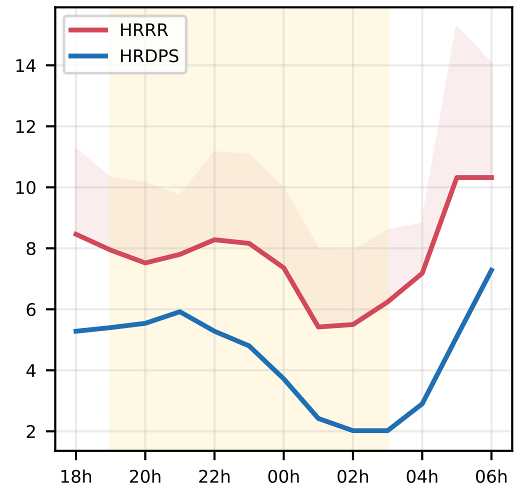
Friday 2026-07-17, 19:00 EDT start · ~03:00 Sat finish · ~40 nm SE across Massachusetts & Cape Cod Bays

NOAA HRRR (3 km) and Environment Canada HRDPS (2.5 km) — two independent high-resolution models.

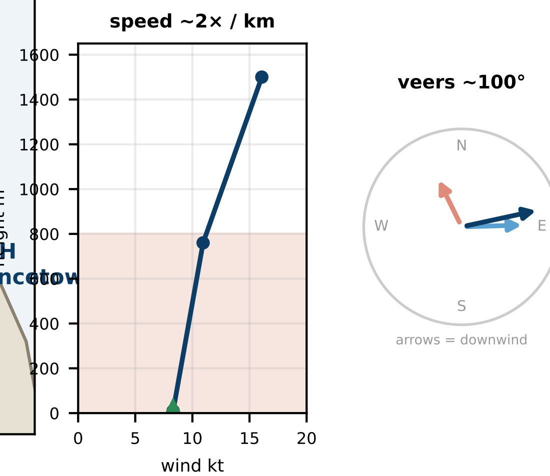
They agree on the overnight but split on the start — see the difference map + tactics on p.2. · 2026-07-16 22:37Z · ~24 h lead.

Route & overnight wind (HRRR mean arrows)

Course-mean wind (kt)



Wind shear (HRRR, mid-bay)



Overnight synopsis

Light, weak-gradient southerly night (flat ~1016 mb, no front). The models AGREE on the overnight: the wind fills S/SW 8-12 kt offshore, a starboard-tack reach that gradually frees toward Race Point, seas ≤2 ft. They SHARPLY DISAGREE on the start: NOAA HRRR has SE ~10 kt (a reach/beat down the rhumb) while Canada HRDPS shows a NW land-breeze ~7 kt (offshore), dying to near-calm by ~01:00 before the southerly fills — opposite first-leg angles. HRDPS has verified better locally this season, but at a post-sunset transition neither is reliable: sail the breeze that's actually on the water, then get offshore into the steadier southerly. Hour-by-hour, tactics, tide and hazards overleaf.

Wind shear — what's happening

After sunset the sea-cooled surface decouples from the wind above — HRRR shows a low-level inversion (the air ~0.8 km up is +1.2°C WARMER than the surface). The course feels a light thermal breeze; aloft the true gradient, W–WNW 13–16 kt, flows almost freely — the two are ~100° apart and speed nearly doubles in the first km. A puff is a brief down-mix of that faster, veered air; by dawn the layer re-mixes and the surface veers to W/SW.
→ what to DO with it: tactics bullet ⑤ (p.2).

Conditions at a glance

Gusts — steady, modest

HRRR gust factor ~1.4–1.5, peaks ~15–16 kt over ~9 kt mean. CAPE 0 + only ~11–15 kt aloft → no convective or momentum-driven gusts. HRDPS: no gust field. Plan smooth, not puffy.

Sea state — benign

~0.2 m / 0.7 ft short-period wind chop (NWS seas ≤1–2 ft). SST ~21°C / 69°F. Flat water — the limiter is wind, not waves.

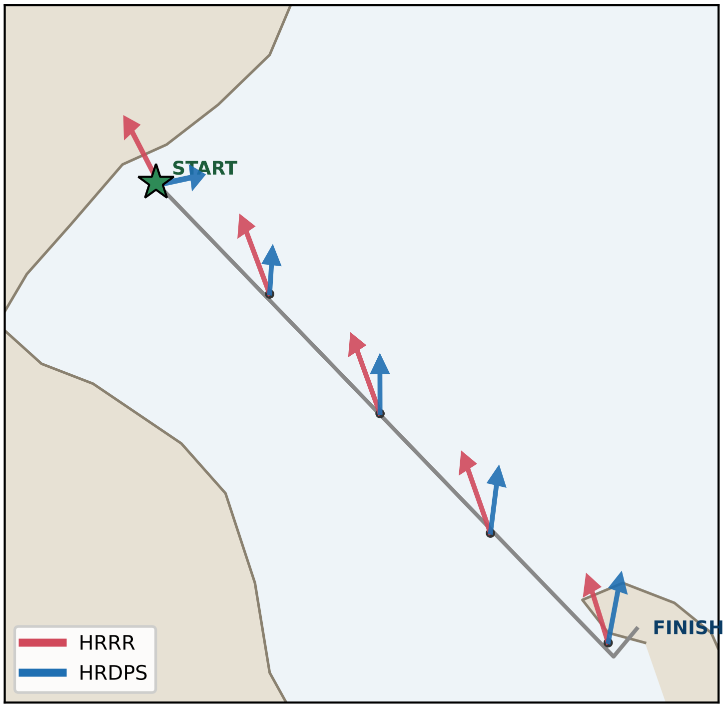
Temperature — mild & near-steady

~18°C / 64–65°F all night on the water (sea-moderated: ~1°F drop offshore, up to ~10°F near the P-town shore). Warm water under cooler air → damp; reinforces the fog watch. Layers for damp, not cold.

Tactics, Strategy & Timing

Marblehead → Provincetown · Fri 2026-07-17 overnight

Where the models differ — mean-wind arrows (length ∝ speed)



HRRR vs HRDPS — mean wind by point

Start (Tinkers Rk)
HRRR 145°/7kt · HRDPS 261°/5kt → Δdir 116°

Off Nahant
HRRR 153°/8kt · HRDPS 185°/2kt → Δdir 32°

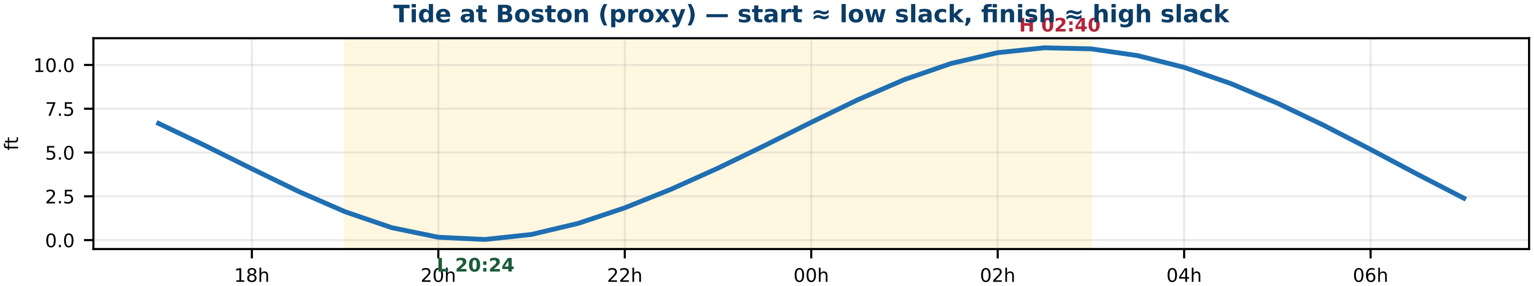
Mid Mass Bay
HRRR 154°/8kt · HRDPS 180°/4kt → Δdir 26°

Stellwagen edge
HRRR 155°/8kt · HRDPS 190°/5kt → Δdir 35°

Race Pt / P-town
HRRR 157°/6kt · HRDPS 194°/6kt → Δdir 37°

Biggest split at the START (near-opposite: a dying SE sea-breeze vs a NW land-breeze). They converge to a common S/SW by Race Point, HRDPS lighter throughout. → The first leg is the uncertain part — sail the actual breeze; for the rest of the night both models agree.

Tide at Boston (proxy) — start ≈ low slack, finish ≈ high slack



Tactics & strategy

① Start (19:00–21:00) — models disagree, so read the water

HRRR: SE ~10 kt (reach/close-hauled). HRDPS: NW land-breeze ~7 kt (offshore), dying to calm ~01:00. Opposite angles — don't pre-commit. Sail the actual dock breeze; whatever it is it's LIGHT (≤8 kt). Get clear air off the line, then work OFFSHORE — the Marblehead shore is the light trap.

② The crossing (21:00–01:00) — reach in the filling southerly

Both models fill S/SW 8-12 kt offshore — a starboard reach that frees as the wind veers right. Pressure lives offshore/south (mid-bay ~10 kt vs shore 2-6). Commit south; stay on the pressure side, don't over-stand north toward Cape Ann.

③ ⚠ LNG Northeast Gateway — hard avoid

Two STL buoys at ~42.40 N (−70.60/−70.59) sit just N of the direct line. NO ENTRY within 500 m, no anchoring within ~1 nm. The rhumb passes ~5 nm south — staying on/south of the line is both faster (more breeze) and legal.

④ Race Point & finish (01:00–03:00) — the tide gate

S/SSW 8-11 kt, gusts to 15 into Cape Cod Bay. Flood builds to Boston high 02:45 ≈ finish. Round Race Point BEFORE ~02:45 to carry the flood in; later boats meet the building ebb. Finish current itself is minor (near high slack).

⑤ Wind shear — shifty, veered puffs; trim for twist

Light SE/S surface under a W–WNW 13–16 kt flow (inversion = decoupled; diagram + explainer on p.1). Puffs arrive VEERED toward W/NW and a bit stronger (cap ~16 kt): upwind, starboard lifts / port headers — tack on the headers, don't commit a side. Trim for TWIST — ease the upper leech (the masthead sees the stronger, veered wind). By dawn the surface veers/builds to W/SW.

⑥ Night & hazards

Dark — thin crescent sets early, so nav by instruments, watches + lights set. Fog watch: July S-flow over ~20°C water; NWS not flagging Fri night but recheck AM. Seas ≤2 ft, no SCA expected.

Confidence: moderate on the overnight reach + tide timing (models agree); low on the exact start (transition + lead). Re-run race morning + pre-start.